Loss function – Cost Function

Loss Function is the **error for ONE data point**

Cost Function is the **average error for ALL data points**

Loss function – Cost Function

Example: Linear Regression

$$Loss_i = (y_{i,actual} - y_{i,predict})^2$$

$$Cost = \frac{Loss_1 + Loss_2 + \cdots + Loss_n}{n}$$

x	y_actual	y_predicted	Loss
1	2	1.5	$(2-1.5)^2 = 0.25$
4	4	3.4	$(4-3.4)^2 = 0.36$
3	3	3.9	$(3-3.9)^2 = 0.81$

Cost = (0.25 + 0.36 + 0.81)/3 = 0.47

Example of Loss function:

1) **Mean Squared Error** - MSE (Used in Linear regression):

$$Loss = (y - \hat{y})^2$$

It is used in house price prediction, salary prediction, etc.

2) **Mean Absolute Error** –MAE (Used in Linear regression):

$$Loss = |y - \hat{y}|$$

Where:

- y = actual value
- $\hat{y}$ = prediction

Example of Loss function:

1) **Binary Cross Entropy** (Used in classification):
Here y is 0 or 1.

$$Loss = -(y\log(\hat{y}) + (1-y)\log(1-\hat{y}))$$

2) **Hinge Loss** (Used in classification by SVM):
Here y is 0 or 1.

$$Loss = max(0, 1 - y \cdot \hat{y})$$

Where:

- y = actual value
- $\hat{y}$ = prediction

Example of Loss function:

Categorical Cross Entropy Loss:

Used in multi-class classification, i.e. there are **more than 2 classes**.

For example, in digit recognition (0 – 9), animal classification (dog-cat-snake)

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

Where:

- C = number of classes
- y_i = actual label (one-hot encoded)
- $\hat{y}_i$ = predicted probability

Example of Loss function:

Huber Loss (Regression):
It combines advantages of **MSE + MAE**.

$$L = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta(|y - \hat{y}| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

Where:

- y = actual value
- $\hat{y}$ = prediction
- δ = threshold

Example of Loss function:

KL Divergence Loss:

Measures difference between **probability distributions.**

It is used in, VAEs, Generative AI, Knowledge distillation

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

Where:

- P = true distribution
- Q = predicted distribution

It essentially tells you how much information is lost when using Q instead of P.

Example of Loss function:

Focal Loss:

It is used for **imbalanced classification problems**.

Example

- Fraud detection
- Cancer detection

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

Where:

- p_t = predicted probability of correct class
- α = class weight
- γ = focusing parameter

Example of Loss function:

Dice Loss (Image Segmentation):

Measures overlap between prediction and ground truth.

$$Dice = \frac{2|A \cap B|}{|A| + |B|}$$

$$Loss = 1 - Dice$$

Where:

- A = predicted pixels
- B = actual pixels

STOP